

HW4

24 октября 2025 г. 23:23

7. Suppose that during the most recent fiscal year, the annual revenue of Acme Computer was 138 billion dollars and its net profit was 8 billion dollars, the annual revenue of Nadir Software was 87 billion dollars and its net profit was 5 billion dollars, and the annual revenue of Quixote Media was 111 billion dollars and its net profit was 13 billion dollars. Determine the truth value of each of these propositions for the most recent fiscal year.
- Quixote Media had the largest annual revenue.
 - Nadir Software had the lowest net profit and Acme Computer had the largest annual revenue.
 - Acme Computer had the largest net profit or Quixote Media had the largest net profit.
 - If Quixote Media had the smallest net profit, then Acme Computer had the largest annual revenue.
 - Nadir Software had the smallest net profit if and only if Acme Computer had the largest annual revenue.

11. Let p and q be the propositions

p : It is below freezing.

q : It is snowing.

Write these propositions using p and q and logical connectives (including negations).

- It is below freezing and snowing.
- It is below freezing but not snowing.
- It is not below freezing and it is not snowing.
- It is either snowing or below freezing (or both).
- If it is below freezing, it is also snowing.
- Either it is below freezing or it is snowing, but it is not snowing if it is below freezing.
- That it is below freezing is necessary and sufficient for it to be snowing.

Problem 1

- False, Because the highest annual income was earned by Acme Compute
- The first and second parts are correct, so this is true
- The first part is false and the second part is true, the condition is 'OR', which means this statement is true.
- The hypothesis of this conditional statement is false and the conclusion is true, so by the truth-table definition this is a true statement.
- Both parts of this biconditional statement are true, so by the truth-table definition this is a true statement.

Problem 2

- $p \wedge q$
- $p \wedge \neg q$
- $\neg p \wedge \neg q$
- $p \vee q$
- $p \rightarrow q$
- $(p \vee q) \wedge (p \rightarrow \neg q)$
- $p \Leftrightarrow q$

29. How many rows appear in a truth table for each of these compound propositions?

- a) $p \rightarrow \neg p$
 b) $(p \vee \neg r) \wedge (q \vee \neg s)$
 c) $q \vee p \vee \neg s \vee \neg r \vee \neg t \vee u$
 d) $(p \wedge r \wedge t) \leftrightarrow (q \wedge t)$

Problem 3

- A) $2^1 = 2$
 B) $2^4 = 16$
 C) $2^6 = 64$
 D) $2^4 = 16$

31. Construct a truth table for each of these compound propositions.

- a) $p \wedge \neg p$ b) $p \vee \neg p$
 c) $(p \vee \neg q) \rightarrow q$ d) $(p \vee q) \rightarrow (p \wedge q)$
 e) $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$
 f) $(p \rightarrow q) \rightarrow (q \rightarrow p)$

Problem 4

a)	p	$\neg p$	$p \wedge \neg p$	b)	$p \vee \neg p$
	<u>T</u>	<u>F</u>	<u>F</u>		<u>T</u>
	F	T	F		T

c)	p	q	$\neg q$	$p \vee \neg q$	$(p \vee \neg q) \rightarrow q$
	<u>T</u>	<u>T</u>	<u>F</u>	<u>T</u>	<u>T</u>
	T	F	T	T	F
	F	T	F	F	T
					F

F F T T T

$$\neg(p \wedge \neg \neg q) \vee q \equiv \neg p \wedge q \vee q \equiv q$$

$$n = 2^2 = 4$$

d)

p	q	$p \vee q$	$p \wedge q$	$(p \vee q) \rightarrow (p \wedge q)$
T	T	T	T	T
T	F	T	F	F
F	T	T	F	F
F	F	F	F	T

e)

p	q	$p \rightarrow q$	$\neg q \rightarrow \neg p$	$(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$
T	T	T	T	T
T	F	F	F	T
F	T	T	T	T
F	F	T	T	T

f)

p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \rightarrow (q \rightarrow p)$
T	T	T	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	F	T

$\begin{array}{cc} \neg & T \\ \neg & F \end{array}$
 $\begin{array}{c} \neg \\ T \end{array}$
 $\neg$

17. When three professors are seated in a restaurant, the hostess asks them: "Does everyone want coffee?" The first professor says: "I do not know." The second professor then says: "I do not know." Finally, the third professor says: "No, not everyone wants coffee." The hostess comes back and gives coffee to the professors who want it. How did she figure out who wanted coffee?

Problem 5

If the first professor did not want coffee, he would have said no, not everyone wants it, therefore he wants coffee, and the same applies to the second professor. Conclusion: the first and second professors want coffee, but the third does not.

7. Use De Morgan's laws to find the negation of each of the following statements.
- a) Jan is rich and happy.
 - b) Carlos will bicycle or run tomorrow.
 - c) Mei walks or takes the bus to class.
 - d) Ibrahim is smart and hard working.

Problem 6

- A) Jan is not rich or not happy
- B) Carlos will not bicycle and not run tomorrow
- C) Mei doesn't walk and doesn't take the bus to class
- D) Ibrahim is not smart and not hard working

9. Show that each of these conditional statements is a tautology by using truth tables.

- a) $(p \wedge q) \rightarrow p$ b) $p \rightarrow (p \vee q)$
 c) $\neg p \rightarrow (p \rightarrow q)$ d) $(p \wedge q) \rightarrow (p \rightarrow q)$
 e) $\neg(p \rightarrow q) \rightarrow p$ f) $\neg(p \rightarrow q) \rightarrow \neg q$

Problem 7

a) $(p \wedge q) \rightarrow p$

p	q	$p \wedge q$	$(p \wedge q) \rightarrow p$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

b) $p \rightarrow (p \vee q)$

p	q	$p \vee q$	$p \rightarrow (p \vee q)$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	T

c) $\neg p \rightarrow (p \rightarrow q)$

p	q	$\neg p$	$p \rightarrow q$	$\neg p \rightarrow (p \rightarrow q)$
T	T	F	T	T
T	F	F	F	T
F	T	T	T	T
F	F	T	T	T

d) $(p \wedge q) \rightarrow (p \rightarrow q)$

p	q	$p \wedge q$	$p \rightarrow q$	$(p \wedge q) \rightarrow (p \rightarrow q)$
T	T	T	T	T
T	F	F	F	T
F	T	F	T	T
F	F	F	T	T

e) $\neg(p \rightarrow q) \rightarrow p$

p	q	$p \rightarrow q$	$\neg(p \rightarrow q)$	$\neg(p \rightarrow q) \rightarrow p$
T	T	T	F	T
T	F	F	T	T
F	T	T	F	T
F	F	T	F	T

f) $\neg(p \rightarrow q) \rightarrow \neg q$

p	q	$p \rightarrow q$	$\neg(p \rightarrow q)$	$\neg q$	$\neg(p \rightarrow q) \rightarrow \neg q$
T	T	T	F	F	T
T	F	F	T	T	T
F	T	T	F	F	T
F	F	T	F	T	T

T	T	T
T	F	F
F	T	T
F	F	T

T
F
F

T
T
T

F
T

T
T
T

Problem 8

11. Show that each conditional statement in Exercise 9 is a tautology without using truth tables.

$$\begin{array}{l|l}
 \text{a) } (p \wedge q) \rightarrow p & \text{b) } p \rightarrow (p \vee q) \\
 (p \vee \neg q) \vee p & \neg p \vee (p \vee q) \\
 & q \vee (\neg p \vee p) \\
 & q \vee T
 \end{array}$$

$(\neg p \vee p) \vee q$ $T \vee q$ T	T
$c) \neg p \rightarrow (p \rightarrow q)$ $p \vee (p \rightarrow q)$ $p \vee (\neg p \vee q)$ $q \vee (\neg p \vee p)$ $q \vee T$ T	$d) (p \wedge q) \rightarrow (p \rightarrow q)$ $\neg(p \wedge q) \vee (\neg p \vee q)$ $(\neg p \vee \neg q) \vee (\neg p \vee q)$ $(\neg p \vee \neg p) \vee (\neg q \vee q)$ $\neg p \vee T$ T
$e) \neg(p \rightarrow q) \rightarrow p$ $\neg(\neg p \vee q) \rightarrow p$ $(p \wedge \neg q) \rightarrow p$ $(\neg p \vee q) \vee p$ $(\neg p \vee p) \vee q$ $T \vee q$ T	$f) \neg(p \rightarrow q) \rightarrow \neg q$ $\neg(\neg p \vee q) \rightarrow \neg q$ $(p \wedge \neg q) \rightarrow \neg q$ $\neg(p \wedge \neg q) \vee \neg q$ $(\neg p \vee q) \vee \neg q$ $(\neg q \vee q) \vee \neg p$ $T \vee \neg p$ T

23. Show that $(p \rightarrow r) \wedge (q \rightarrow r)$ and $(p \vee q) \rightarrow r$ are logically equivalent.

Problem 9

P	Q	r	$P \rightarrow r$	$Q \rightarrow r$	$(P \rightarrow r) \wedge (Q \rightarrow r)$	$P \vee Q$	$(P \vee Q) \rightarrow r$
T	T	T	T	T	T	T	T
T	T	F	F	F	F	T	F
T	F	T	T	T	T	T	T
T	F	F	F	T	F	T	F
F	T	T	T	F	F	T	F
F	T	F	T	T	T	T	T
F	F	T	T	T	T	F	F
F	F	F	T	T	T	F	F

9. Let $P(x)$ be the statement “ x can speak Russian” and let $Q(x)$ be the statement “ x knows the computer language C++.” Express each of these sentences in terms of $P(x)$, $Q(x)$, quantifiers, and logical connectives. The domain for quantifiers consists of all students at your school.

- There is a student at your school who can speak Russian and who knows C++.
- There is a student at your school who can speak Russian but who doesn't know C++.
- Every student at your school either can speak Russian or knows C++.
- No student at your school can speak Russian or knows C++.

Problem 10

- $\exists x(P(x) \wedge Q(x))$
- $\exists x(P(x) \wedge \neg Q(x))$
- $\forall x(P(x) \vee Q(x))$
- $\forall x \neg (P(x) \vee Q(x))$

15. Determine the truth value of each of these statements if the domain for all variables consists of all integers.

- a) $\forall n(n^2 \geq 0)$ b) $\exists n(n^2 = 2)$
 c) $\forall n(n^2 \geq n)$ d) $\exists n(n^2 < 0)$

A) $2^2 = 4 \geq 0$ - true; $(-2)^2 = 4 \geq 0$ - true; $0^2 = 0 \geq 0$ - true; Result = TRUE
 B) $0^2 = 0$ - false; $1^2 = 1$ - false; $2^2 = 4$ - false; Result = FALSE
 C) $0^2 = 0 \geq 0$ - true; $1^2 = 1 \geq 1$ - true; $(-2)^2 = 4 \geq 4$ - true; Result = TRUE
 D) $(-1)^2 = 1 < 0$ - false; Result = FALSE

Problem 11

17. Suppose that the domain of the propositional function $P(x)$ consists of the integers 0, 1, 2, 3, and 4. Write out each of these propositions using disjunctions, conjunctions, and negations.

- a) $\exists x P(x)$ b) $\forall x P(x)$ c) $\exists x \neg P(x)$
 d) $\forall x \neg P(x)$ e) $\neg \exists x P(x)$ f) $\neg \forall x P(x)$

- a) $\exists x P(x) = P(0) \vee P(1) \vee P(2) \vee P(3) \vee P(4)$
 b) $\forall x P(x) = P(0) \wedge P(1) \wedge P(2) \wedge P(3) \wedge P(4)$
 c) $\exists x \neg P(x) = \neg P(0) \vee \neg P(1) \vee \neg P(2) \vee \neg P(3) \vee \neg P(4)$
 d) $\forall x \neg P(x) = \neg P(0) \wedge \neg P(1) \wedge \neg P(2) \wedge \neg P(3) \wedge \neg P(4)$
 e) $\neg \exists x P(x) = \neg(P(0) \vee P(1) \vee P(2) \vee P(3) \vee P(4))$
 f) $\neg \forall x P(x) = \neg(P(0) \wedge P(1) \wedge P(2) \wedge P(3) \wedge P(4))$

Problem 12

23. Translate in two ways each of these statements into logical expressions using predicates, quantifiers, and logical connectives. First, let the domain consist of the students in your class and second, let it consist of all people.

- a) Someone in your class can speak Hindi.
- b) Everyone in your class is friendly.
- c) There is a person in your class who was not born in California.
- d) A student in your class has been in a movie.
- e) No student in your class has taken a course in logic programming.

Problem 13

a) $G(x)$ = «x can speak Hindi»; $C(x)$ = «x is a student in your class»

All people: $\exists x(C(x) \wedge G(x))$; Student in your class: $\exists xG(x)$

b) $G(x)$ = «x is freindly»; $C(x)$ = «x is a student in your class»

All people: $\forall x(C(x) \rightarrow G(x))$; Student in your class: $\forall xG(x)$

c) $G(x)$ = «x born in California»; $C(x)$ = «x is a student in your class»

All people: $\exists x(C(x) \wedge \neg G(x))$; Student in your class: $\exists x\neg G(x)$

d) $G(x)$ = «x has been a movie»; $C(x)$ = «x is a student in your class»

All people: $\exists x(C(x) \wedge G(x))$; Student in your class: $\exists xG(x)$

e) $G(x)$ = «x take a course in logic programming»; $C(x)$ = «x is a student in your class»

All people: $\neg \exists x(C(x) \wedge G(x))$; Student in your class: $\neg \exists xG(x)$

13. Let $M(x, y)$ be “x has sent y an e-mail message” and $T(x, y)$ be “x has telephoned y,” where the domain consists of all students in your class. Use quantifiers to express each of these statements. (Assume that all e-mail messages that were sent are received, which is not the

way things often work.)

- a) Chou has never sent an e-mail message to Koko.
- b) Arlene has never sent an e-mail message to or telephoned Sarah.
- c) José has never received an e-mail message from Deborah.
- d) Every student in your class has sent an e-mail message to Ken.
- e) No one in your class has telephoned Nina.
- f) Everyone in your class has either telephoned Avi or sent him an e-mail message.
- g) There is a student in your class who has sent everyone else in your class an e-mail message.
- h) There is someone in your class who has either sent an e-mail message or telephoned everyone else in your class.
- i) There are two different students in your class who have sent each other e-mail messages.
- j) There is a student who has sent himself or herself an e-mail message.
- k) There is a student in your class who has not received an e-mail message from anyone else in the class and who has not been called by any other student in the class.
- l) Every student in the class has either received an e-mail message or received a telephone call from another student in the class.
- m) There are at least two students in your class such that one student has sent the other e-mail and the second student has telephoned the first student.
- n) There are two different students in your class who between them have sent an e-mail message to or telephoned everyone else in the class.

Problem 14

- a) $\neg M(\text{Chou}, \text{Koko})$
- b) $\neg (M(\text{Arlene}, \text{Sarah}) \vee T(\text{Arlene}, \text{Sarah}))$
- c) $\neg M(\text{Deborah}, \text{José})$
- d) $\forall x M(x, \text{Ken})$
- e) $\neg \exists x T(x, \text{Nina})$
- f) $\forall x (T(x, \text{Avi}) \vee M(x, \text{Avi}))$
- g) $\exists x \forall y (y \neq x \rightarrow M(x, y))$
- h) $\exists x \forall y (y \neq x \rightarrow (M(x, y) \vee T(x, y)))$
- i) $\exists x \exists y (x \neq y \wedge M(x, y) \wedge M(y, x))$
- j) $\exists x M(x, x)$
- k) $\exists x \forall y (y \neq x \rightarrow (\neg M(y, x) \wedge \neg T(y, x)))$
- l) $\forall x \exists y (y \neq x \wedge (M(y, x) \vee T(y, x)))$
- m) $\exists x \exists y (x \neq y \wedge M(x, y) \wedge T(y, x))$
- n) $\exists x \exists y (x \neq y \wedge \forall z (z \neq x \wedge z \neq y \rightarrow (M(x, z) \vee T(x, z) \vee M(y, z) \vee T(y, z))))$